

```

3 (*we list all four examples in one .wl file for the convenience of demonstratio
4 (*However, we recommend each example in one file.*)

In[1]:= (*using same way of loading as AMflow and Blade*)
8 current = If[$FrontEnd===Null,$InputFileName,NotebookFileName[]]//DirectoryName
9 Get[current<>"../.."<>"Baikov.wl"];

BaikovAll: a package for generating Baikov representations for a given Feynman integra
Version: 1.1.0
If you come across any problem, please contact phamapku14@gmail.com

```

Outer-Massive double box

```

In[3]:= dlist= {
17     PD[k1,0,-m2],
18     PD[k1,-p1,-m2],
19     PD[k1,-p1-p2,-m2],
20     PD[k1-k2,0,0],
21     PD[k2,-p1-p2,-m2],
22     PD[k2,-p1-p2-p3,-m2],
23     PD[k2,0,-m2],
24     PD[k2,-p1,-m2],
25     PD[k1,-p1-p2-p3,-m2]
26 };(*input of propagators and independent scalar products*)
27 conservation={p4→-p1-p2-p3};(*if dependent variables does not appear, this can
28 kinematics={
29     SProd[p1]→0,
30     SProd[p2]→0,
31     SProd[p3]→0,
32     SProd[p4]→0,
33     SProd[p1+p2]→s,
34     SProd[p2+p3]→t
35 };(*define independent scalar products, SProd[a]=SProd[a,a]*)

In[6]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
39 krep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all s

In[8]:= stdrep=BaikovRep[dlist,{k1,k2},{p1,p2,p3},Abst→True];(*the standard representa
43 resultmat=AllSectorBaikovMat[stdrep,krep,Exc→{}];(*derive all representations

The global variable $LoopNum has been set to 2
The global variable $BvNum has been set to 9

In[10]:= Export[current<>"db_rep.wxf",resultmat,PerformanceGoal→"Size"];

49 (*we can now explore some function in this family*)

```

```
In[11]:= (*1. Get the loop by loop representation from the whole representations*)
53 (*now we need to specify the sector we want, {1,2,3,4,5,6,7} specifies which pr
54 (*two independent loop-by-loop representation will be extracted*)
55 toprep=GetBaikovMatRep[resultmat,{1,2,3,4,5,6,7}]
```

```
Out[11]= {{ {x9}, {{ {G[{p1, p2, p3}, {p1, p2, p3}], ε}, {-G[{k2, p1, p2}, {k2, p1, p2}],
-1 + 1/2 (2 + 2 ε)}, {G[{k2, p1, p2, p3}, {k2, p1, p2, p3}], 1/2 + 1/2 (-2 - 2 ε)},
{G[{k1, k2, p1, p2}, {k1, k2, p1, p2}], 1/2 + 1/2 (-2 - 2 ε)}}, 1/(64 π^3 Gamma[1/2 - ε]^2)}},
{{ {x8}, {{ {G[{p1, p2, p3}, {p1, p2, p3}], ε},
{-G[{k1, p1 + p2, p3}, {k1, p1 + p2, p3}], -1 + 1/2 (2 + 2 ε)},
{G[{k1, p1 + p2, p2, p3}, {k1, p1 + p2, p2, p3}], 1/2 + 1/2 (-2 - 2 ε)},
{G[{k1, k2, p1 + p2, p3}, {k1, k2, p1 + p2, p3}], 1/2 + 1/2 (-2 - 2 ε)}},
1/(64 π^3 Gamma[1/2 - ε]^2)}}}}
```

```
In[12]:= (*2. get the explicit expressions for the representation*)
59 (*we can also get the dimension of the top topology under maximal cut*)
60 (*dim means the number of master integrals under maximal cut without considerin
61 toprepexp = Times@@Power@@@toprep[[1,2,1]]//Gram2Poly[#,krep]& //Simplify
62 dim = GetDimension[toprepexp/.{Subscript[x, i_/(i<8)]>0},{Subscript[x, 8]}]
```

```
Out[12]= 16^{1+ε} (-s t (s + t))^ε (s (m2 s + x5 (x7 - x8) + (s - x7) x8 + x8^2))^ε
(-4 m2 s^2 t - 4 m2 s t^2 + s^2 t^2 + t^2 x5^2 + s^2 x6^2 -
2 s t^2 x7 + t^2 x7^2 - 2 t x5 (-s x6 + (2 s + t) x7 + s (t - x8)) -
2 s^2 t x8 + 2 s t x7 x8 + s^2 x8^2 - 2 s x6 (s t - t x7 + (s + 2 t) x8))^{-1/2-ε}
(-4 m2 s^2 x4 + s^2 x4^2 + 4 m2 s x3 x7 - 2 s x3 x4 x7 - 4 m2 s x5 x7 + x3^2 x7^2 +
x2^2 (s (-4 m2 + s) + x5^2 - 2 s x7 + x7^2 - 2 x5 (s + x7)) + x1^2 (x5 - x8)^2 - 4 m2 s x3 x8 -
2 s^2 x4 x8 + 2 s x3 x4 x8 + 4 m2 s x5 x8 + 4 m2 s x7 x8 + 2 s x3 x7 x8 - 2 x3^2 x7 x8 - 4 m2 s x8^2 +
s^2 x8^2 - 2 s x3 x8^2 + x3^2 x8^2 + 2 x2 (x3 (2 m2 s - x7^2 + (s - x5) x8 + x7 (s + x5 + x8)) + s
(-2 m2 x7 + x4 (-s + x5 + x7 - 2 x8) + 4 m2 x8 - s x8 + x7 x8 + x5 (-2 m2 - 2 x7 + x8))) +
2 x1 (s (x5 - x8) (2 m2 - x4 + x8) - x3 (2 m2 s + x5 (x7 - x8) + (2 s - x7) x8 + x8^2) +
x2 (2 m2 s - x5^2 + (s - x7) x8 + x5 (s + x7 + x8)))^{-1/2-ε}
```

```
Out[13]= 4
```

```

In[14]:= (*3. considering the symmetry of top sector by analyzing the polynomial in star
66 poly=stdrep[1,2,1]//Gram2Poly[#,krep]&;
67 symrule=PolySym[poly,Join[Subscript[x,#]&/@Range[$BvNum],{s,t,m2}],$BvNum]

```

```

Out[15]= {{Cycles[{{2, 9}, {6, 8}}], Cycles[{{1, 2}, {3, 9}, {5, 6}, {7, 8}, {10, 11}}],
Cycles[{{1, 2, 3, 9}, {5, 6, 7, 8}, {10, 11}}],
Cycles[{{1, 3}, {5, 7}}], Cycles[{{1, 3}, {2, 9}, {5, 7}, {6, 8}}],
Cycles[{{1, 5}, {2, 6}, {3, 7}, {8, 9}}],
Cycles[{{1, 5}, {2, 8}, {3, 7}, {6, 9}}],
Cycles[{{1, 6}, {2, 5}, {3, 8}, {7, 9}, {10, 11}}],
Cycles[{{1, 6, 3, 8}, {2, 7, 9, 5}, {10, 11}}],
Cycles[{{1, 7}, {2, 6}, {3, 5}, {8, 9}}],
Cycles[{{1, 7}, {2, 8}, {3, 5}, {6, 9}}],
Cycles[{{1, 8, 3, 6}, {2, 5, 9, 7}, {10, 11}}],
Cycles[{{1, 8}, {2, 7}, {3, 6}, {5, 9}, {10, 11}}],
Cycles[{{1, 9, 3, 2}, {5, 8, 7, 6}, {10, 11}}],
Cycles[{{1, 9}, {2, 3}, {5, 8}, {6, 7}, {10, 11}}]},
{{X2 -> X9, X6 -> X8, X8 -> X6, X9 -> X2},
{X1 -> X2, X2 -> X1, X3 -> X9, X5 -> X6, X6 -> X5, X7 -> X8, X8 -> X7, X9 -> X3, s -> t, t -> s},
{X1 -> X9, X2 -> X1, X3 -> X2, X5 -> X8, X6 -> X5, X7 -> X6, X8 -> X7, X9 -> X3, s -> t, t -> s},
{X1 -> X3, X3 -> X1, X5 -> X7, X7 -> X5},
{X1 -> X3, X2 -> X9, X3 -> X1, X5 -> X7, X6 -> X8, X7 -> X5, X8 -> X6, X9 -> X2},
{X1 -> X5, X2 -> X6, X3 -> X7, X5 -> X1, X6 -> X2, X7 -> X3, X8 -> X9, X9 -> X8},
{X1 -> X5, X2 -> X8, X3 -> X7, X5 -> X1, X6 -> X9, X7 -> X3, X8 -> X2, X9 -> X6},
{X1 -> X6, X2 -> X5, X3 -> X8, X5 -> X2, X6 -> X1, X7 -> X9, X8 -> X3, X9 -> X7, s -> t, t -> s},
{X1 -> X8, X2 -> X5, X3 -> X6, X5 -> X9, X6 -> X1, X7 -> X2, X8 -> X3, X9 -> X7, s -> t, t -> s},
{X1 -> X7, X2 -> X6, X3 -> X5, X5 -> X3, X6 -> X2, X7 -> X1, X8 -> X9, X9 -> X8},
{X1 -> X7, X2 -> X8, X3 -> X5, X5 -> X3, X6 -> X9, X7 -> X1, X8 -> X2, X9 -> X6},
{X1 -> X6, X2 -> X7, X3 -> X8, X5 -> X2, X6 -> X3, X7 -> X9, X8 -> X1, X9 -> X5, s -> t, t -> s},
{X1 -> X8, X2 -> X7, X3 -> X6, X5 -> X9, X6 -> X3, X7 -> X2, X8 -> X1, X9 -> X5, s -> t, t -> s},
{X1 -> X2, X2 -> X3, X3 -> X9, X5 -> X6, X6 -> X7, X7 -> X8, X8 -> X5, X9 -> X1, s -> t, t -> s},
{X1 -> X9, X2 -> X3, X3 -> X2, X5 -> X8, X6 -> X7, X7 -> X6, X8 -> X5, X9 -> X1, s -> t, t -> s}}}

```

```

In[16]:= (*check these are indeed the symmetry of this polynomial*)
71 PolySymCheck[poly,symrule[[2]]]

```

```

Out[16]= {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}

```

```

In[17]:= (*4. get all zero sectors derived from Baikov representation*)
75 zerolist=GetMatZeroSector[resultmat,$BvNum,{8,9}] (*the last list is where the
76 PadRight[Reverse[IntegerDigits[#,2]], $BvNum]&/@zerolist (*look up these sector

```

```

Out[17]= {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21,
22, 23, 24, 32, 33, 34, 35, 36, 37, 38, 39, 40, 48, 49, 50, 51, 52, 53, 54,
55, 56, 64, 65, 66, 67, 68, 69, 70, 71, 72, 80, 81, 82, 83, 84, 86, 88, 96,
97, 98, 99, 100, 101, 102, 103, 104, 112, 113, 114, 115, 116, 118, 120}

```

```

Out[18]= {{0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0, 0, 0, 0, 0},
{1, 1, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 0, 0, 0, 0, 0, 0}, {1, 0, 1, 0, 0, 0, 0, 0, 0, 0},
{0, 1, 1, 0, 0, 0, 0, 0, 0, 0}, {1, 1, 1, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 1, 0, 0, 0, 0, 0, 0},
{1, 0, 0, 1, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 1, 0, 0, 0, 0, 0, 0}, {1, 1, 0, 1, 0, 0, 0, 0, 0, 0},
{0, 0, 1, 1, 0, 0, 0, 0, 0, 0}, {1, 0, 1, 1, 0, 0, 0, 0, 0, 0}, {0, 1, 1, 1, 0, 0, 0, 0, 0, 0},
{1, 1, 1, 1, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 1, 0, 0, 0, 0, 0},
{0, 1, 0, 0, 1, 0, 0, 0, 0, 0}, {1, 1, 0, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 1, 0, 0, 0, 0, 0},
{1, 0, 1, 0, 1, 0, 0, 0, 0, 0}, {0, 1, 1, 0, 1, 0, 0, 0, 0, 0}, {1, 1, 1, 0, 1, 0, 0, 0, 0, 0},
{0, 0, 0, 1, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 1, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 1, 0, 0, 0, 0},
{0, 1, 0, 0, 0, 1, 0, 0, 0, 0}, {1, 1, 0, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 1, 0, 0, 1, 0, 0, 0, 0},
{1, 0, 1, 0, 0, 1, 0, 0, 0, 0}, {0, 1, 1, 0, 0, 1, 0, 0, 0, 0}, {1, 1, 1, 0, 0, 1, 0, 0, 0, 0},
{0, 0, 0, 1, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 1, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 1, 1, 0, 0, 0},
{0, 1, 0, 0, 0, 1, 1, 0, 0, 0}, {1, 1, 0, 0, 0, 1, 1, 0, 0, 0}, {0, 0, 1, 0, 0, 1, 1, 0, 0, 0},
{1, 0, 1, 0, 0, 1, 1, 0, 0, 0}, {0, 1, 1, 0, 0, 1, 1, 0, 0, 0}, {1, 1, 1, 0, 0, 1, 1, 0, 0, 0},
{0, 0, 0, 1, 1, 1, 0, 0, 0, 0}, {0, 1, 0, 0, 1, 1, 1, 0, 0, 0}, {1, 1, 0, 0, 1, 1, 1, 0, 0, 0},
{0, 0, 1, 0, 1, 1, 1, 0, 0, 0}, {1, 0, 1, 0, 1, 1, 1, 0, 0, 0}, {0, 1, 1, 0, 1, 1, 1, 0, 0, 0},
{1, 1, 1, 0, 1, 1, 1, 0, 0, 0}, {0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0}, {0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0},
{1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0}, {0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0}, {1, 1, 0, 0, 1, 1, 1, 1, 0, 0, 0},
{0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0}, {0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 0}, {0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0}}

```

Linear propagator involved

```

In[19]:= (*now we present above example with linear propagator involved*)
84  (*just replace PD with LPD: LPD[k,p,-m2]=k.p-m2, PD[k,p,-m2]=(k+p)^2-m2*)
85  dlist={
86      PD[k1,0,-m2],
87      LPD[k1,-p1,-m2],
88      LPD[k1,-p1-p2,-m2],
89      LPD[k1,k2,0],
90      PD[k2,-p1-p2,-m2],
91      PD[k2,-p1-p2-p3,-m2],
92      PD[k2,0,-m2],
93      LPD[k2,-p1,-m2],
94      PD[k1,-p1-p2-p3,-m2]
95  };(*input of propagators and independent scalar products*)
96  conservation={p4→-p1-p2-p3};(*if dependent variables does not appear, this can
97  kinematics={
98      SProd[p1]→0,
99      SProd[p2]→0,
100     SProd[p3]→0,
101     SProd[p4]→0,
102     SProd[p1+p2]→s,
103     SProd[p2+p3]→t
104  };(*define independent scalar products, SProd[a]=SProd[a,a]*)

In[22]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
108  nkrep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all

```

```
In[24]:= (*one can compare the result with the former example*)
112 nkrep(*new one with linear propagator*)
113 krep(*former one with all normal propagators*)
```

```
Out[24]= {k1 · k1 → m2 + x1, k1 · k2 → x4, k1 · p1 → -m2 - x2, k1 · p2 → x2 - x3,
          k1 · p3 →  $\frac{1}{2} (2 m^2 + x_1 + 2 x_3 - x_9)$ , k2 · k2 → m2 + x7, k2 · p1 → -m2 - x8,
          k2 · p2 →  $\frac{1}{2} (2 m^2 + s - x_5 + x_7 + 2 x_8)$ , k2 · p3 →  $\frac{1}{2} (-s + x_5 - x_6)$ , p1 · p1 → 0,
          p1 · p2 →  $\frac{s}{2}$ , p1 · p3 →  $-\frac{s}{2} - \frac{t}{2}$ , p2 · p2 → 0, p2 · p3 →  $\frac{t}{2}$ , p3 · p3 → 0}
```

```
Out[25]= {k1 · k1 → m2 + x1, k1 · k2 →  $\frac{1}{2} (2 m^2 + x_1 - x_4 + x_7)$ , k1 · p1 →  $\frac{1}{2} (x_1 - x_2)$ ,
          k1 · p2 →  $\frac{1}{2} (s + x_2 - x_3)$ , k1 · p3 →  $\frac{1}{2} (-s + x_3 - x_9)$ , k2 · k2 → m2 + x7,
          k2 · p1 →  $\frac{1}{2} (x_7 - x_8)$ , k2 · p2 →  $\frac{1}{2} (s - x_5 + x_8)$ , k2 · p3 →  $\frac{1}{2} (-s + x_5 - x_6)$ ,
          p1 · p1 → 0, p1 · p2 →  $\frac{s}{2}$ , p1 · p3 →  $-\frac{s}{2} - \frac{t}{2}$ , p2 · p2 → 0, p2 · p3 →  $\frac{t}{2}$ , p3 · p3 → 0}
```

```
In[26]:= (*derive the representations in the same way*)
117 stdrep=BaikovRep[dlist,{k1,k2},{p1,p2,p3},Abst→True];(*the standard representation*)
118 nresultmat=AllSectorBaikovMat[stdrep,nkrep,Exc→{}];(*derive all representation*)
```

The global variable \$LoopNum has been set to 2
The global variable \$BvNum has been set to 9

additional parameters in definition of propagators

```
In[28]:= (*in some cases, there may be parameters between the combinations of momenta*)
126 (*we need to declare these variables as numbers*)
127 DeclareVarAsNum[{r,z}]
128 dlist={
129     PD[l1, -(1 - z)*k1, 0],
130     PD[l1, z*k1, 0],
131     PD[l2, 0, 0],
132     PD[l3, 0, 0],
133     PD[l1 - l2, -(1 - z)*k1, 0],
134     PD[l1 + l3, z*k1, 0],
135     PD[l1 + l3, z*k1 + k2, 0],
136     PD[l1 - l2 + l3, z*k1 + k2, 0],
137     PD[l1 - l2, z*k1 + k2, 0],
138     PD[l2, k1, 0],
139     PD[l3, k1, 0],
140     PD[l1, k2, 0]
141 };(*input of propagators and independent scalar products*)
142 conservation={};(*if dependent variables does not appear, this can be an empty
143 kinematics={
144     SProd[k1]→0,
145     SProd[k2]→0,
146     SProd[k1,k2]→s/2
147 };(*define independent scalar products, SProd[a]=SProd[a,a]*)
```

variables: {r, z} have been declared as numbers.

```
In[32]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
151 krep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all s
```

```
In[34]:= krep
```

```
Out[34]= {k1 · l1 →  $\frac{1}{2} (-x_1 + x_2)$ , k1 · l2 →  $\frac{1}{2} (-x_3 + x_{10})$ ,
k1 · l3 →  $\frac{1}{2} (-x_4 + x_{11})$ , k2 · l1 →  $\frac{1}{2} (-z x_1 - x_2 + z x_2 + x_{12})$ ,
k2 · l2 →  $\frac{1}{2} (s z - x_1 - z x_1 + z x_2 + x_3 + x_5 - x_9 - x_{10} + x_{12})$ ,
k2 · l3 →  $\frac{1}{2} (-s z + z x_1 + x_2 - z x_2 - x_6 + x_7 - x_{12})$ , l1 · l1 →  $z x_1 + x_2 - z x_2$ ,
l1 · l2 →  $\frac{1}{2} (x_1 + z x_3 - x_5 + x_{10} - z x_{10})$ , l1 · l3 →  $\frac{1}{2} (-x_2 - x_4 + z x_4 + x_6 - z x_{11})$ ,
l2 · l2 →  $x_3$ , l2 · l3 →  $\frac{1}{2} (-s z + z x_1 - z x_2 + x_7 - x_8 + x_9 - x_{12})$ ,
l3 · l3 →  $x_4$ , k1 · k1 → 0, k1 · k2 →  $\frac{s}{2}$ , k2 · k2 → 0}
```

```
In[35]:= stdrep=BaikovRep[dlist,{l1,l2,l3},{k1,k2},Abst→True];(*the standard representa
158 resultmat=AllSectorBaikovMat[stdrep,krep];(*derive all representations from the
```

The global variable \$LoopNum has been set to 3
The global variable \$BvNum has been set to 12

```
In[37]:= Export[current<>"para_rep.wxf",resultmat,PerformanceGoal→"Size");
```

```
In[38]:= Options[GetBaikovMatRep]
```

```
Out[38]:= {ExcVar → {}, looporder → 3, ForceAdd → 0, Baikov`Private`debug → False}
```

```
In[39]:= GetBaikovMatRep[resultmat,{1,2,3,4,5,6}](*this is a zero sector as we can see s
```

... **GetBaikovMatRep::zero** : This is a zero sector with the following variables being integrated out:

{x₁₂, x₁₁, x₉, x₈, x₇}

```
Out[39]= {{ {x12, x11, x9, x8, x7},
  {{ {G[{l1, -l1 + l2, k1}, {l1, -l1 + l2, k1}], 1 +  $\frac{1}{2}(-2 - 2\epsilon)$ },
    {-G[{l1 + k1 z}, {l1 + k1 z}], -2 +  $\frac{1}{2}(2 + 2\epsilon)$ },
    {G[{l1 + k1 z, l3}, {l1 + k1 z, l3}],  $\frac{3}{2} + \frac{1}{2}(-2 - 2\epsilon)$ },
    {-G[{k1}, {k1}], -2 +  $\frac{1}{2}(2 + 2\epsilon)$ }}, 0}},
  {{ {x11, x10, x9, x8, x7}, {{ {G[{k1, k2}, {k1, k2}],  $\frac{1}{2}(-1 + 2\epsilon)$ },
    {G[{k2 + l1 + k1 z, k1 + k2, k2}, {k2 + l1 + k1 z, k1 + k2, k2}], 1 +  $\frac{1}{2}(-2 - 2\epsilon)$ },
    {-G[{-k1 + l1 + k1 z}, {-k1 + l1 + k1 z}], -2 +  $\frac{1}{2}(2 + 2\epsilon)$ },
    {G[{-k1 + l1 + k1 z, l2}, {-k1 + l1 + k1 z, l2}],  $\frac{3}{2} + \frac{1}{2}(-2 - 2\epsilon)$ },
    {-G[{l1 + k1 z}, {l1 + k1 z}], -2 +  $\frac{1}{2}(2 + 2\epsilon)$ },
    {G[{l1 + k1 z, l3}, {l1 + k1 z, l3}],  $\frac{3}{2} + \frac{1}{2}(-2 - 2\epsilon)$ }},
     $\frac{(-1)^{-\epsilon} 2^{-3-2\epsilon}}{\pi^{5/2} \Gamma[2 - 2\epsilon] \Gamma[\frac{3}{2} - \epsilon]}$  } } } }
```


five-loop banana: Derive representation starting from somewhere not being the standard representation

```

In[40]:= dlist= {
175     PD[k1,0,-m2],
176     PD[k2-k1,0,0],
177     PD[k3-k2,0,0],
178     PD[k4-k3,0,0],
179     PD[k5-k4,0,0],
180     PD[k5,p,-m2],
181     PD[k2,0,-m2],
182     PD[k3,0,-m2],
183     PD[k4,0,-m2],
184     PD[k5,0,-m2],
185     PD[k1,p,-m2],
186     PD[k2,p,-m2],
187     PD[k3,p,-m2],
188     PD[k4,p,-m2],
189     PD[k5-k1,0,0],
190     PD[k5-k2,0,0],
191     PD[k5-k3,0,0],
192     PD[k4-k1,0,0],
193     PD[k4-k2,0,0],
194     PD[k3-k1,0,0]
195 };(*input of propagators and independent scalar products*)
196 conservation={};(*if dependent variables does not appear, this can be an empty
197 kinematics={
198     SProd[p]→s
199 };(*define independent scalar products, SProd[a]=SProd[a,a]*)

In[43]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
203 krep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all s

In[45]:= (*we use Exc→{...} to first specify which variables are not integrated out.*)
207 (*Then we get all possible loop-by-loop representations for the top sector*)
208 (*at last, we start from these loop-by-loop representations and derive all the
209 AbsoluteTiming[
210 stdrep=BaikovRep[dlist,{k1,k2,k3,k4,k5},{p},Abst→True];
211 resultmat=AllSectorBaikovMat[stdrep,krep,Exc→{Subscript[x,1],Subscript[x,2]},
212 resultc=AllSectorBaikovMatC[resultmat,Subscript[x,#]&/@Range[20],krep];
213 Export[current<>"banana5l_rep.wxf",resultc,PerformanceGoal→"Size"];
214 ]

```

The global variable \$LoopNum has been set to 5

The global variable \$BvNum has been set to 20

Out[45]= {477.233, Null}

```

In[46]:= (*there are totally 42 representations for this single top topology*)
218      (*but they are symmetric as we will see*)
219      toprep=GetBaikovMatRep[resultc,{1,2,3,4,5,6}];
220      toprep//Length

```

Out[47]= 42

```

In[48]:= (*half in integer power, half in half-integer power*)
224      toprep[[1]]//Simplify
225      toprep[[-1]]//Simplify
226

```

Out[48]= $\left\{ \{x_{20}, x_{19}, x_{18}, x_{17}, x_{16}, x_{15}, x_{14}, x_{13}, x_{12}, x_{11}\}, \right.$
 $\left\{ \left\{ G[\{p\}, \{p\}], -1 + \epsilon, \{-G[\{k_2\}, \{k_2\}], -1 + \epsilon, \left\{ G[\{k_1, k_2\}, \{k_1, k_2\}], \frac{1}{2} - \epsilon \right\}, \right. \right.$
 $\left. \{-G[\{k_3\}, \{k_3\}], -1 + \epsilon, \left\{ G[\{k_2, k_3\}, \{k_2, k_3\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \{-G[\{k_4\}, \{k_4\}], -1 + \epsilon, \left\{ G[\{k_3, k_4\}, \{k_3, k_4\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \{-G[\{k_5\}, \{k_5\}], -1 + \epsilon, \left\{ G[\{k_5, p\}, \{k_5, p\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \left\{ G[\{k_4, k_5\}, \{k_4, k_5\}], \frac{1}{2} - \epsilon \right\} \right\}, \frac{(-1)^{-2\epsilon}}{32 \pi^{5/2} \Gamma\left[\frac{3}{2} - \epsilon\right]^5} \right\}$

Out[49]= $\left\{ \{x_{19}, x_{17}, x_{16}, x_{14}, x_{13}, x_{12}, x_{10}, x_9, x_8, x_7\}, \right.$
 $\left\{ \left\{ G[\{p\}, \{p\}], -1 + \epsilon, \{-G[\{k_1 - k_3\}, \{k_1 - k_3\}], -1 + \epsilon, \right. \right.$
 $\left. \left\{ G[\{k_1 - k_3, k_2 - k_3\}, \{k_1 - k_3, k_2 - k_3\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \{-G[\{k_1 - k_4\}, \{k_1 - k_4\}], -1 + \epsilon, \left\{ G[\{k_1 - k_4, k_3 - k_4\}, \{k_1 - k_4, k_3 - k_4\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \{-G[\{k_1 - k_5\}, \{k_1 - k_5\}], -1 + \epsilon, \left\{ G[\{k_1 - k_5, k_4 - k_5\}, \{k_1 - k_5, k_4 - k_5\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \{-G[\{k_1 + p\}, \{k_1 + p\}], -1 + \epsilon, \left\{ G[\{k_1 + p, p\}, \{k_1 + p, p\}], \frac{1}{2} - \epsilon \right\}, \right.$
 $\left. \left\{ G[\{k_1 + p, k_5 + p\}, \{k_1 + p, k_5 + p\}], \frac{1}{2} - \epsilon \right\} \right\}, \frac{(-1)^{-2\epsilon}}{32 \pi^{5/2} \Gamma\left[\frac{3}{2} - \epsilon\right]^5} \right\}$